

ABSOLUTE GRAVITY

Nagarkot Geodetic Observatory, Nepal

March/April 1991

Observations, corrections and results.

Gravity ties to Kathmandu and Simira airports.

Dan Winester, Jack Fried and Brent Bernard

National Geodetic Survey, Rockville Md

Laxman Shrestha, Buddhi N. Shrestha and Gajanan Adiga

HMG Survey Department, Dilli Bazar, Nepal

Roger Bilham and Jim Faller

University of Colorado, Boulder, CO, 80309

GEODETIC SURVEY BRANCH
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ABSOLUTE GRAVITY, Nagarkot, Nepal 1991

NGS Rockville Md:

Survey of Nepal:

Coordinated by:

Dan Winester, Jack Fried and Brent Bernard

Laxman Shrestha and Gajanan Adiga

Roger Bilham, Jim Faller and Buddhi N. Shrestha

Summary of measurements

The purpose of measuring absolute gravity in the Himalaya was to establish a reference datum for the local gravity network in Nepal and to establish points that may be remeasured to reveal changes of elevation in future years. The original plan was to measure absolute gravity at three locations: in the Greater Himalaya, in the Lesser Himalaya and in the Terai bordering the northern plains of India. Each absolute gravity point was scheduled to be co-located with a GPS control point so that an independent estimate of vertical deformation might be possible.

The plan we adopted differed in three ways from the above:

1) One absolute-g site only was measured at Nagarkot (FAGS-1). The corrected value of the FAGS-1 indoor point at ground level for the period 3/30/91-4/2/91 is $978494834.7 \pm 6.7 \mu\text{gal}$. The gravity gradient at floor level (zero to 0.43m) was $4.4194 \mu\text{gal/cm}$.

2) Relative ties were made to three GPS points: Nagarkot, Kathmandu airport and Simira Airport. The relative differences from FAGS-1 to these points are listed on the next page.

The ties were undertaken using a pair of Model D LaCoste Romberg meters. For Nagarkot the GPS point is less than 10 m from the brick building where GPS measurements were made. The Kathmandu Airport tie was undertaken using road transport (multiple ties over the 33-km-long 1.5 hour road linking Nagarkot to the capital). The Simira tie was made by flying several times between Simira and Kathmandu. The Model D gravimeter has just sufficient range to accommodate the gravity variation associated with the vertical change in height between Nagarkot and Kathmandu, and also the latitude change and vertical range combination between Kathmandu and Simira.

3) The limited number of sites suitable for gravity measurements has resulted in no gravity measurements at points suspected to be rising in the Greater and Lesser Himalaya. Simira is south of the Lesser Himalaya and Kathmandu and Nagarkot lie between the Lesser and the Greater Himalaya. Future Model D or Model G gravimeter ties be made from Kathmandu airport to GPS points elsewhere in Nepal are needed to correct this limitation in the 1991 measurements.

A removal truck was used to meet the several hundred pounds of equipment from the plane and to store the packaging at Nagarkot. The power at Nagarkot was found to be unreliable for the gravity measurements as was the portable generator used to provide backup power. Measurements for this reason were spread over a longer period than is usual. Air conditioning was requested for the gravimeter but was found to be unnecessary in Nagarkot. A decision to occupy only one point "absolutely" and the other points using Model D gravimeters was made because:

- a) the absolute gravimeter was damaged in transit to Kathmandu or on the road to Nagarkot and might have further been damaged by additional road transport.
- b) suitable temperature control from air conditioners was unavailable at the other selected sites, and an air conditioner would have had to have been trucked in from India together with a 15 kw generator.
- c) Power outages at Nagarkot reduced the time available for measurements at additional sites.

The new gravity base stations provide a framework for the local Nepal gravity network. It is anticipated that future gravity measurements will extend this network throughout the country. The absolute accuracy of the 1991 measurements is $\pm 6 \mu\text{gals}$ or approximately $\pm 1.5 \text{ cm}$ in elevation.

Funding support for the measurements was provided by NASA grant NAGW-2704. A description of the JILA absolute gravimeter follows the observational data.



UNITED STATES DEPARTMENT OF COMMERCE
National Oceanic and Atmospheric Administration
NATIONAL OCEAN SERVICE
~~OFFICE OF CHARTING AND GEODETIC SERVICES~~
ROCKVILLE, MARYLAND 20852

Coast and Geodetic Survey
11 June 1991

Dr. Roger Bilham
CIRES, Univ. of Colorado
Boulder, CO 80309

Dear Roger:

Enclosed are gravity base station descriptions for occupied sites in Nepal. A copy of these will be sent to Buddhi Shrestha. The NAGARKOT FAGS-1 absolute gravity value will be available from Dr. Peter. The gradients at NAGARKOT FAGS-1 from floor to 83 cm is 0.44194 mgal/m and from floor to 120 cm is 0.43923 mgal/m. Relative to the floor value at NAGARKOT FAGS-1 at the following gravity transfers:

NAGARKOT GPS	- 0.691 $\pm$ 0.002 mgals
KATHMANDU J	+166.469 $\pm$ 0.005
SIMARA J	+368.599 $\pm$ 0.017
SIMARA GPS	+368.706 $\pm$ 0.013

Sincerely,

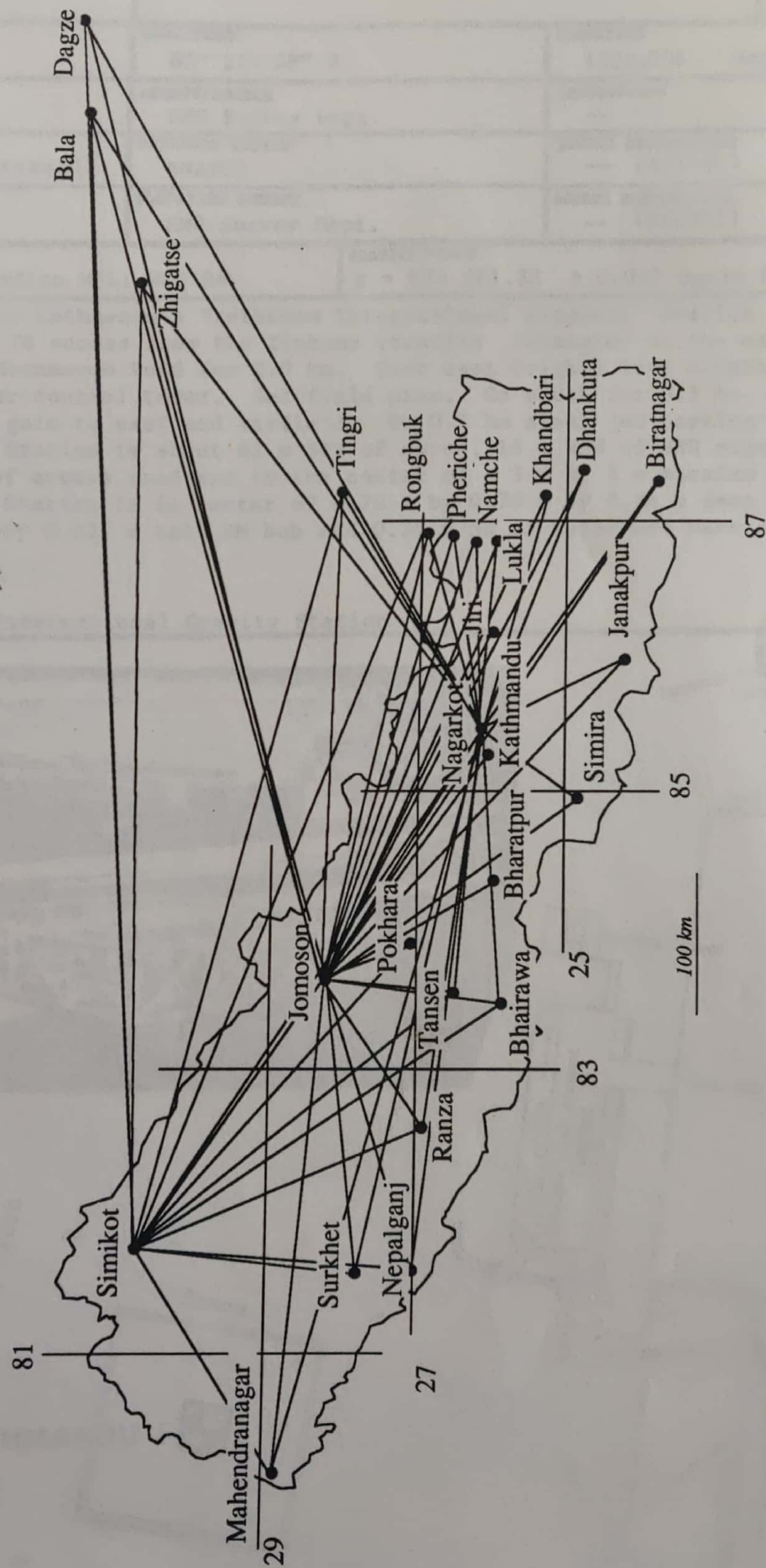
Daniel Winester, Geodesist
National Geodetic Survey, N/CG 161N



NEPAL/TIBET GPS Survey

Direct ties to base stations at Simikot, Jomoson and Nagarkot 25 March-12 April 1991

287



GRAVITY STATION DESCRIPTION		STATION TYPE	STATION DESIGNATION
COUNTRY Nepal		National Reference Base	KATHMANDU J
DISTRICT (ZILA) Kathmandu		ZONE (ANCHAL) Bagmati	CITY Kathmandu
LATITUDE 27° 41' 50" N		LONGITUDE 85° 21' 28" E	ELEVATION 1332.006 meters
GRAVITY STATION MARK Benchmark Hub		AGENCY/SOURCE HMG Survey Dept.	INSCRIPTION --
POSITION REFERENCE GPS Position (unprocessed)		POSITION SOURCE UNAVCO	SOURCE DESIGNATION -- (4/1991)
ELEVATION REFERENCE BM Elevation		ELEVATION SOURCE HMG Survey Dept.	SOURCE DESIGNATION -- (4/1991)

QUAD
027274424

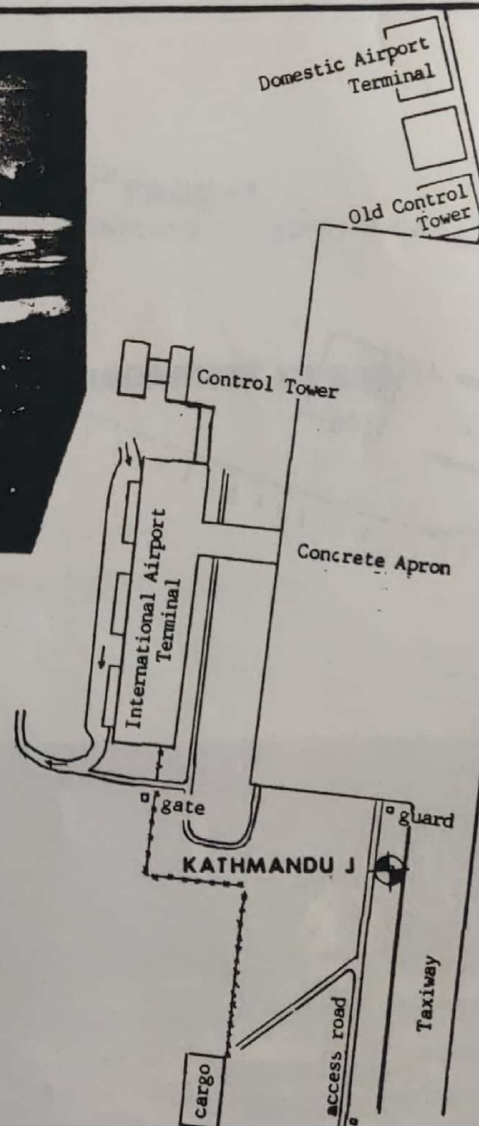
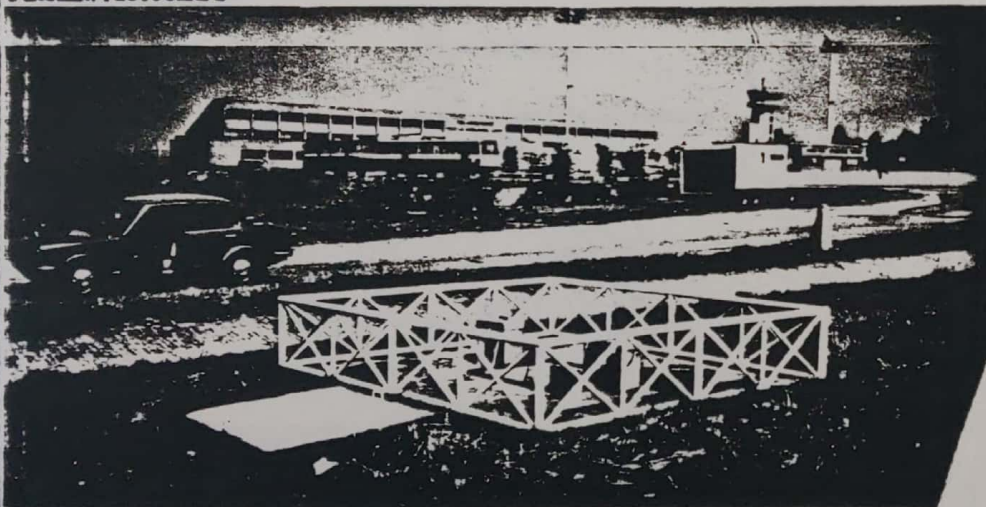
POSITION/ELEVATION REMARKS
1st order levels; Indian MSL; WGS 84

GRAVITY VALUE
g = 978 661.22 ± 0.047 mgals (512 STRE, 1984)

DESCRIPTION Station is at Kathmandu's Trebhuvan International Airport. Station is 3.8 km ESE of the Royal Palace. To access from the Tinkune (traffic triangle) on the east side of Kathmandu, go NNE on Meanmoven Road for 2.0 km. Turn east (right) into airport and go 0.3 km to Pass Office under control tower. Get field pass. Go south for 0.3 km, passed International Terminal, to gate to east and airfield. Go 0.2 km along jet parking apron to access road to SSW (right). Station is about 62 m SSW of apron, 16 m WNW of WNW edge of taxiway, 10.5 m ESE of center of access road and in the center of a 3 m by 3 m macadam area surrounded by a white fence. Station is in center of 0.70 m by 0.70 m by 0.36 m deep concrete pit and over 0.030 m wide by 0.025 m tall BM hub and 0.32 m SW of Reference Mark and 1.5 m ESE of witness sign.

OTHER STATION DESIGNATIONS: International Gravity Station

DIAGRAM/PHOTOGRAPH



DATE OF PHOTO: 5 Apr 1991

DESCRIPTION BY
Bernard/Winester

AGENCY
NOAA/NOS/NGS

DATE
7 April 1991

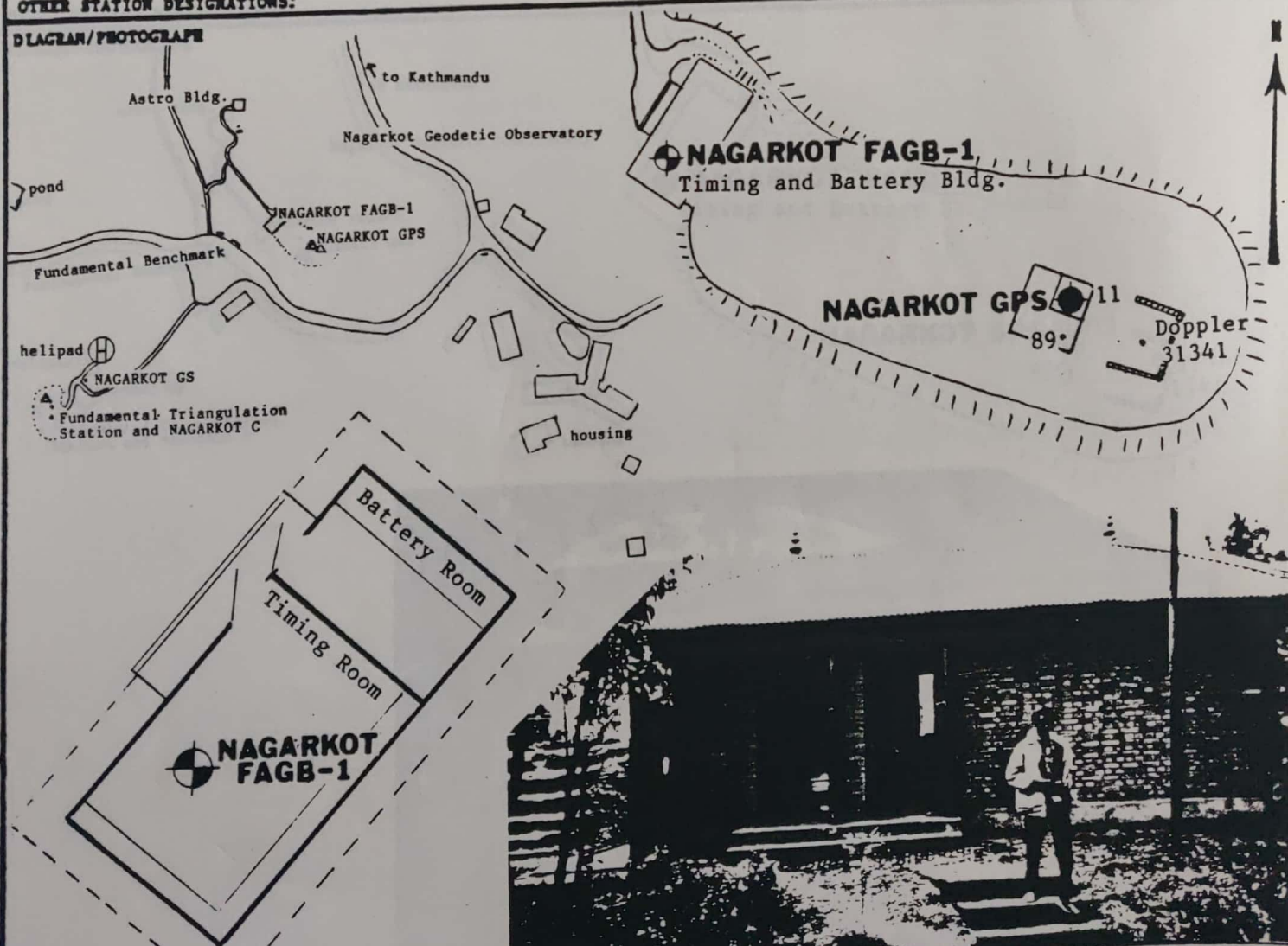
GRAVITY STATION DESCRIPTION		STATION TYPE Absolute Site	STATION DESIGNATION NAGARGOT FAGB-1
COUNTRY Nepal	ZONE (ANCHAL) Bagmati	CITY Nagarkot	
DISTRICT (ZILA) Bhaktapur/Kabhre Palanchok		QUAD 027274134	
LATITUDE 27° 41' 35" N	LONGITUDE 85° 31' 16" E	ELEVATION 2150.564 meters	
GRAVITY STATION MARK 19 mm brass plug	AGENCY/SOURCE NOAA/NGS	INSCRIPTION NFAGB-1 1991	
POSITION REFERENCE Scaled from GPS station	POSITION SOURCE UNAVCO & NOAA/NGS	SOURCE DESIGNATION -- (4/1991)	
ELEVATION REFERENCE Disk Elevation	ELEVATION SOURCE HMG Survey Dept.	SOURCE DESIGNATION -- (4/1991)	

POSITION/ELEVATION REMARKS

1st order levels; Indian MSL; WGS 84

GRAVITY VALUE

DESCRIPTION Station is at HMG Survey Department's Geodetical Observatory - Nagarkot. Station is 19.4 airline km east of the Royal Palace in Kathmandu. To access from the Tinkune (traffic triangle) on east side of Kathmandu, go easterly towards Bhaktapur for 4.7 km. Turn north (left) and go 0.4 km to second turn to east (right). Go easterly up a winding, bumpy road for about 20 km to second guard gate of Nagarkot Army Post. Bear left and go southeast for 2.7 km on dirt road to upper parking lot of Observatory. Station is uphill via footpath to NE in the Timing Room of the Timing and Battery Bldg. (3.7m by 7.5m). Station is 0.93 m SE of NW wall and 2.83 m SW of NE wall of room. Plug is epoxied flush into the thin concrete floor. Contact is Buddhi N. Shrestha, Director General, HMG Survey Dept. at 411-897 in Kathmandu. Site phone is 211-009. FAGB-1 stands for Fundamental Absolute Gravity Base - Number 1.

OTHER STATION DESIGNATIONS:
DIAGRAM/PHOTOGRAPH


DATE OF PHOTO: 3 Apr 1991

DESCRIPTION BY
Bernard/Winester

AGENCY

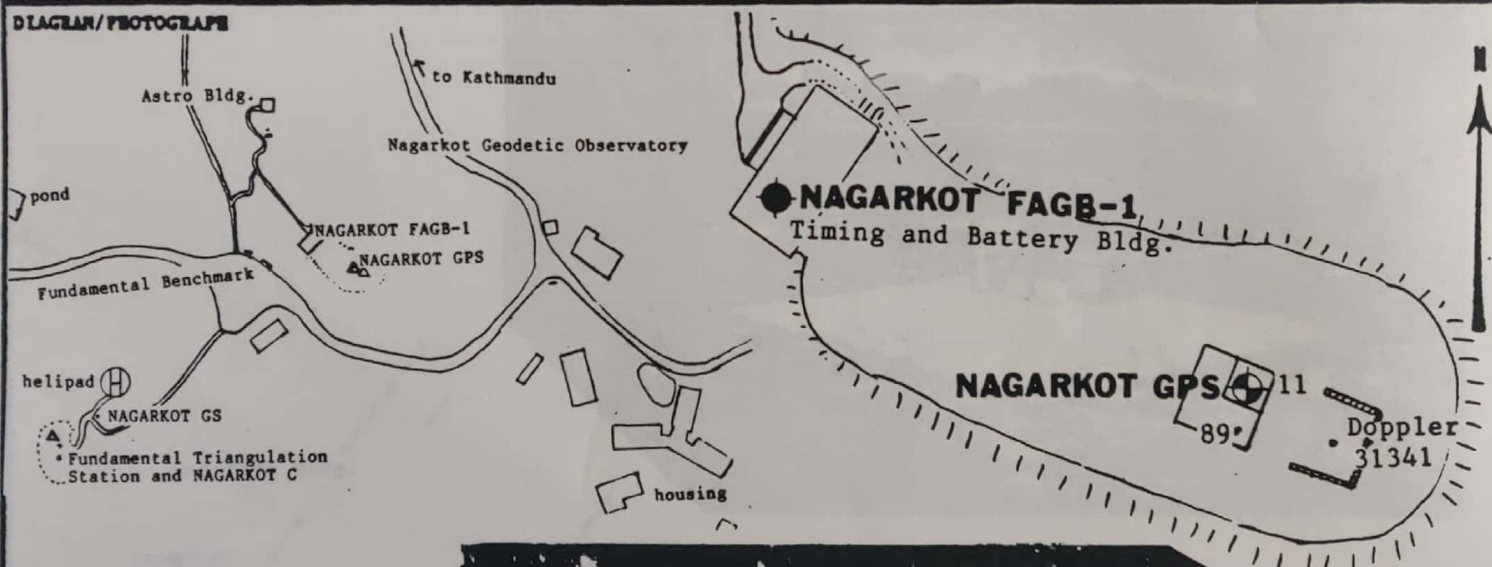
NOAA/NOS/NGS

DATE

7 April 1991

GRAVITY STATION DESCRIPTION		STATION TYPE	STATION DESIGNATION
COUNTRY	Nepal	Absolute Excenter	NAGARKOT GPS
DISTRICT (ZILA)	Bhaktapur/Kabhre Palanchok	ZONE (ANCHAL) Bagmati	CITY Nagarkot
LATITUDE	27° 41' 34" N	LONGITUDE	85° 31' 16" E
GRAVITY STATION MARK	Vertical Rod in Pier	ELEVATION	2152.789 meters
POSITION REFERENCE	GPS position (unprocessed)	AGENCY/SOURCE	UNAVCO
ELEVATION REFERENCE	Disk Elevation	INSCRIPTION	11
POSITION/ELEVATION REMARKS	1st order levels, Indian MSL; WGS-84		
GRAVITY VALUE	027274134		
DESCRIPTION	Station is at HMG Survey Department's Geodetical Observatory - Nagarkot. Station is 19.4 airline km east of the Royal Palace in Kathmandu. To access from the Tinkune (traffic triangle) on east side of Kathmandu, go easterly towards Bhaktapur for 4.7 km. Turn north (left) and go 0.4 km to second turn to east (right). Go easterly up a winding, bumpy road for about 20 km to second guard gate of Nagarkot Army Post. Bear left and go southeast for 2.7 km on dirt road to upper parking lot of Observatory. Station is uphill via footpath to northeast, 18.5 m SE of Battery & Timing Bldg's east corner, 4.2 m NW of Doppler station, 2.0 m NNE of GPS point 89, in east quadrant of 3.3 m squared concrete pad and in center of 1 m square, isolated, concrete pier inscribed GPS Main Station Nagarkot 1991. Pier goes down about 1 m to weathered rock. Rod goes down 0.3 m and then angles to side. Arrow on pier points north.		
OTHER STATION DESIGNATIONS:			

DIAGRAM/PHOTOGRAPH



DATE OF PHOTO: 3 Apr 1991

DESCRIPTION BY
Bernard/Winester

AGENCY
NOAA/NOS/NGS

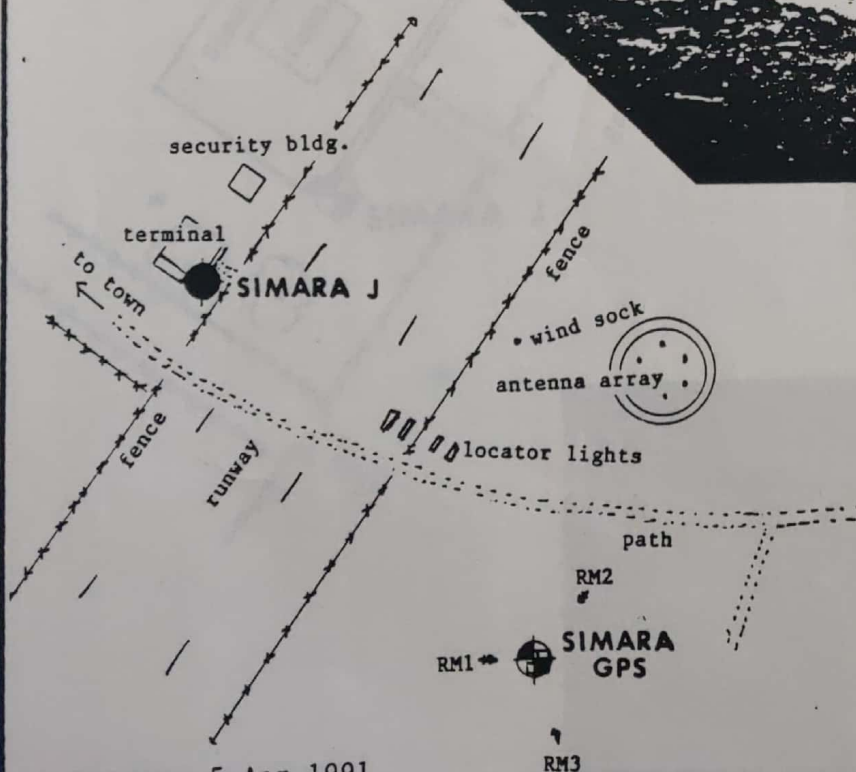
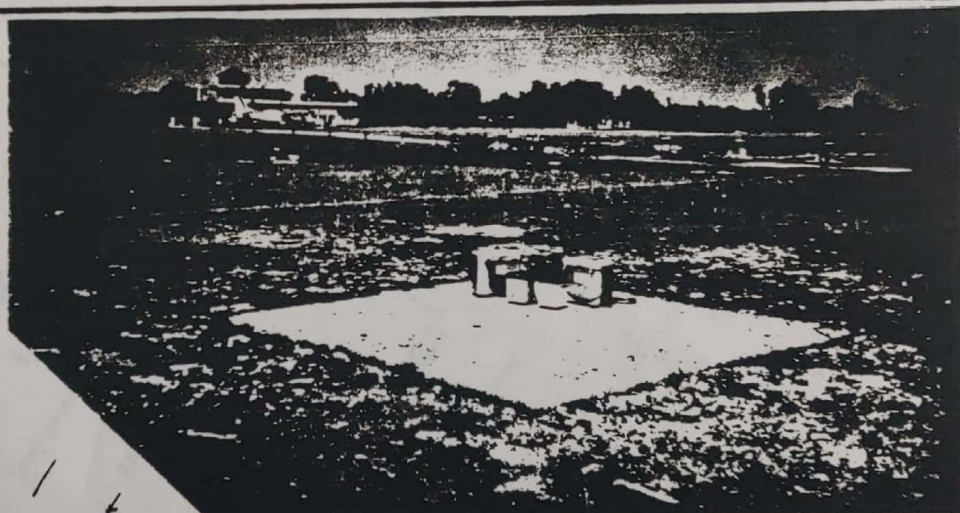
DATE
7 April 1991

GRAVITY STATION DESCRIPTION		STATION TYPE Base Station	STATION DESIGNATION SIMARA GPS
COUNTRY Nepal	ZONE (ANCHAL) Nara Yani	CITY Simara	
DISTRICT (ZILA) Bara			QUAD 027275112
LATITUDE 27° 09' 45" N	LONGITUDE 84° 58' 54" E	ELEVATION 132.5 meters	
GRAVITY STATION MARK Vertical Rod in Pier	AGENCY/SOURCE UNAVCO	INSCRIPTION 34 SIMR	
POSITION REFERENCE GPS Position (unprocessed)	POSITION SOURCE UNAVCO	SOURCE DESIGNATION -- (4/1991)	
ELEVATION REFERENCE Estimated from BM	ELEVATION SOURCE NOAA/NGS & HMG Survey Dept.	SOURCE DESIGNATION --	
POSITION/ELEVATION REMARKS Indian MSL; WGS 84		GRAVITY VALUE	

DESCRIPTION Station is on the Simara Airport grounds, Simara, Nepal. Airport is on east side of Simara and 20 km NNE of Birganj. Station is on SW side of grass runway, near center of old, abandoned east-west runway, 125.3 m S10E of wind sock, 74.8 m S30E of aircraft locator lights, 104 m north of D. Shamser's house, 17.35 m east of RM 1 on old runway marker, 30.68 m S62W of RM 2 on old runway marker and 30.90 m N16W of RM 3 on 0.3 m squared concrete post. Station is in center of 1 m squared concrete pier at NW corner of 3.3 m concrete pad. Pier is 2.0 m deep and belled at bottom and set into soft, sandy soil. Steel rod goes down 0.3 m in concrete and then angles to the side.

OTHER STATION DESIGNATIONS:

DIAGRAM/PHOTOGRAPH



DATE OF PHOTO: 5 Apr 1991

DESCRIPTION BY
Bernard/Winester

AGENCY
NOAA/NOS/NGS

DATE
7 April 1991

GRAVITY STATION DESCRIPTION		STATION TYPE	STATION DESIGNATION
COUNTRY Nepal		Base Station	SIMARA J
DISTRICT (ZILA) Bara		ZONE (ANCHAL) Nari Yani	CITY Simara
LATITUDE 27° 09' 49" N		LONGITUDE 84° 59' 49" E	ELEVATION 131.739 meters
GRAVITY STATION MARK Benchmark Hub		AGENCY/SOURCE HMG Survey Dept.	INSCRIPTION --
POSITION REFERENCE Topo Map 1:50,000		POSITION SOURCE Surveyor General of India	SOURCE DESIGNATION Panchayat 72 ^T 6 (c.1945)
ELEVATION REFERENCE Disk Elevation		ELEVATION SOURCE HMG Survey Dept.	SOURCE DESIGNATION -- (1989)

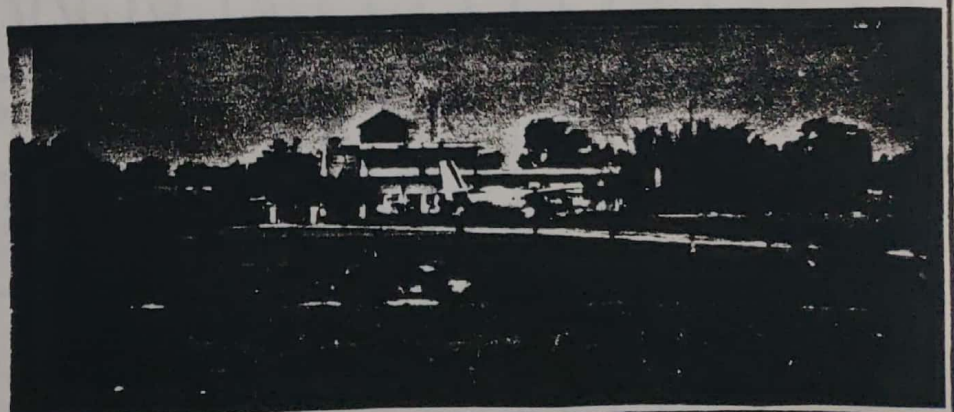
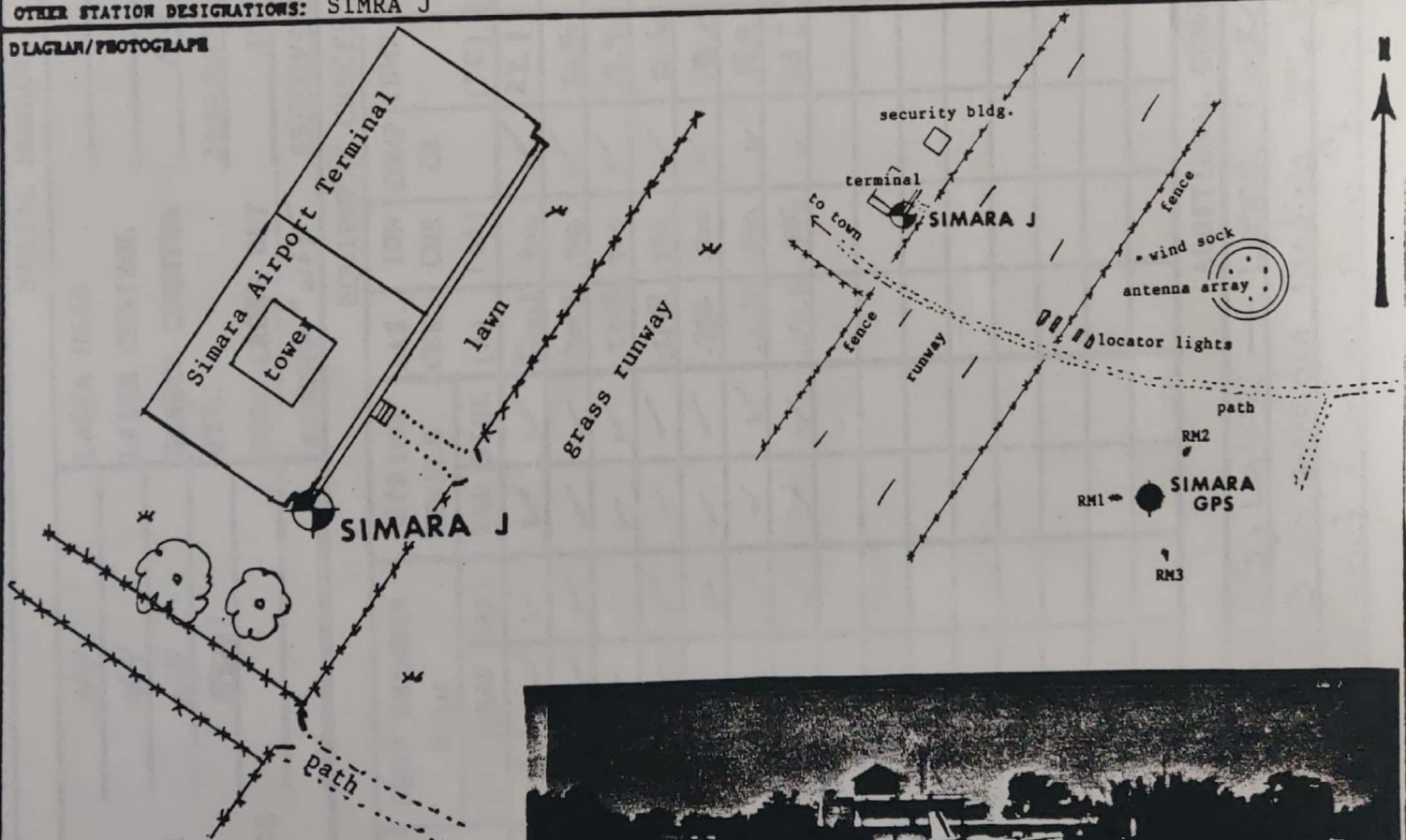
POSITION/ELEVATION REMARKS
1st order levels; Indian MSL; adj. to WGS 84

GRAVITY VALUE
g = 978 863.32 ± 0.070 mgals (512 STRE, 1984)

DESCRIPTION Station is at the Simara Airport Terminal, Simara, Nepal. Airport is on east side of Simara and 20 km NNE of Birganj. Station is at SSE corner (field side) of terminal bldg., over brass hub set into concrete sidewalk at ground level. Station is about 0.7 m away from terminal wall and is below concrete walkway along ESE side of terminal.

OTHER STATION DESIGNATIONS: SIMRA J

DIAGRAM/PHOTOGRAPH



DATE OF PHOTO: 5 Apr 1991

DESCRIPTION BY
Bernard/Winester

AGENCY
NOAA/NOS/NGS

DATE
7 April 1991

JILA #4
ABSOLUTE GRAVITY DETERMINATION

Site: Matmunda AA Start date: 3.3.74 End date: 4.3.74
LAT: 27 41 34 LON: 274 28 43.7 Elevation: 2151 m

Number of drop sequences . . . : 11
Drops per sequence . . . : 250
Sequences using red laser. . . : 10
Sequences using blue laser . . : 7

	GRAVITY (mean)	STD. DEV.
Observed gravity from meter	:	:
Add 3 sigma rejections	2: <u>410.4</u>	:
Add grav. tide program corrections	1: <u>978 494 410.9</u>	:
Add local atmosphere corrections	3: <u>411.8</u>	:
Add synoptic. atm. corrections	:	:
Add ocean loading corrections	4: <u>412.0</u>	:
Add water table correction	:	:
Add polar motion correction	:	:
Add laser drift correction	:	:
Add laser-head temp. correction	:	:
Gravity determination	:	:

Average std. dev. of observation : 11.7 ugals
Difference between means of the two laser settings : 19.4 ugals

Grav. gradient est. by relative meters : 4.4 ugals/cm : -0.1
Weighted mean instrument height . . . : 21.4 cm

Gravity reduced to one meter height : 394.1 : 6.4
Gravity reduced to ground level . . : 978 494 834.7 : ± 6.7

Comments : System response correction was +4.3 ugals.
Left spot on left was used. Interferometer was focused
alignment to Matmunda level. The effect of
a 15 ugals bias due to beam misalignment. It was
added to the floor and give meter values.
Large scatter in set means was due to interferometer alignment
problems.

COMMENTS KATHMANDU, NEPAL
HMG SURVEY NAGARKOT OBSERVATORY
MARCH 30, 1991 1251Z

Nagarkot GPS Site Position ²⁵ km to West

27° 41' 34.24" N = 27.6928N

85 31' 16.30" E = - 85.5212 E

ELEVATION 2131 meters (height above WGS84 ellipsoid)

WHERE to BEGIN? Equipment arrived and took 5 days to clear customs. Dropper/SS/Keithly pallet split up and both chamber cases had bent metal and were laying on side.

Another container had a puncture - you get the picture. 2 1/2 hour ride on pot-holed road to observatory. Pumpdown initiated @ 1900 Local.

Ion pump on @ 0830 Local. 200 uamps. Equip

set up was wrot with problems. #1 optical mount for Pinhole jarred loose and required repair. This went ok although reflected

interferometer spot seems to show more interference by getting larger/fuzzy. This

has been resolved and reflected spot looks good.

#2 Superspring malfunction indicated by

rapid dampening (no Pumpdown) plus an

apparent coupling to floor through support

structure. First investigation was of level bubbles being misadjusted. We could not correct problem

by diddling levels in various combinations in an

attempt to let mass hang freely. So we removed

cannister to look for loose parts when it

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Organization NOAA/NOS/NGS

COMMENTS

became apparent that the springs main stainless steel tube (connected to flexures) had been mistreated. The safety devices which were installed last spring have 4 set screws with a 4mil gap to the steel tube protecting the spring and what it looked like was that the spring had been banded a number of times resulting in a slight bend of the tube so that now it contacted the set screws. We proceeded to back the screws out another 4mils and tested the unit to satisfaction. #3 Power here is a nightmare. Just about anything that's turned on causes the SS Elgar to lose lock on the voltage and frequency. We have overcome this sort of an all out power failure which apparently can occur. We will watch this carefully! #4 Grounding: Building has an open ground. Last night while setting up just about everything I touched gave me a jolt including the Baker Turbo Pump which is ungrounded. Fixed this by stripping the paint on a 2 inch feed pipe for wiring and connected a grounding belt between pipe and Electronic chassis. #5 Radio Telephone! The ^{CHARLOTTEVILLE} ~~GREENSBORO~~ mystery is solved! When Radio telephone transmitting dropper is out of control. To solve this we said no more calls and turned the amplifier off! Nice guysch? #6 Heat? what heat. we have a generator a heater but no adapter to go from a US

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COMMENTS

Extension cord to a German plug. Solution will arrive tomorrow morning from Kathmandu. Building is a very solid brick timbering house on a bare mica schist. 12" concrete floor w/ a metal door and last night the temp only varied 2°C. So for tonight I will check the equip @ 02:00 local to make sure all is OK. #6 All three of us have got the stomach bug but are coping.

March 30, 1991 1300Z First set begun - monitoring drop on scope but every time a drop triggers the Elgar loses lock. Turned scope off. Residuals look very reasonable. # looks good so far. SS looks stable. Vacuum and feedthrough holding up. Cumulative effect of 3 pumpdowns without heat tape is that the average vacuum pressure has risen. Still well within operational limits but will require heat possibly in Hawaii. Wind outside has picked up and we will submit wx records which they collect 50m from the station to submit with the data.

Disturbance seen approximately 2/3rds through first set caused by the Nepalese guard next door opening and closing the heavy metal door which doesn't close smoothly to check if I was still there. Otherwise set looks excellent.

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Organization NOAA/NOS/NGS

COMMENTS

NAGRAKOT Observatory, NEPAL

Adjusted SS levels after Set # 1. Each bubble had drifted almost 1 diameter:

MARCH 31, 1991 0530 Local first 6 sets complete.

Temperature surprisingly stable. Light ~~and~~ RAIN and fog outside. - 0023Z SERIOUS cloud burst overhead w/ 1 mean lightning bolt. NOT EVEN A

brown out of power surprisingly. System ✓ was perfect. Change in Ref. Ht. must be due to bad tape Reading. No adjustment to tripod was made.

Temperature probe Reading 1.5°C higher than calibrated thermometer.

YESTERDAY the Radio transmitter signal was also affecting the thermocouple by causing ERRATIC readings. Correction should be uniform for first 6 sets. We will continue to update.

Gravity readings from last night ARE excellent 4 sigma's for sets 3-6 below 10 microgals.

Set #9 Temp. starting to fluctuate with in observation tolerances (10:49 Local). SKIES starting to CLEAR.

Sets proceeding without incident.

1055 Local - Complete power out - Transferred frequency / Volt meter to TOPAZ's to watch

power drain. Will shut down and save data

and equipment if necessary. UPS voltage @ 1055 117VAC

@ 11:10 113VAC on UPS power to system. 1 UPS FOR

TOR pump only. Well that was it 11:12 Local the

whole system crashed. I turned off everything.

UPS to pump still turning. Got to go.

Next Page

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COMMENTS

Nagrokot Observatory Cont'n

0645Z 1225 Local March 31, 1991

Re-initialized system. Saved DDT Data for set #10 (approx 170 drops) to file called "Nag". System was good but temperature has risen as clouds are breaking. Because of this I believe that the Laser has locked at a lower heater current and rather than cycle it up to .300 amps as we did long ago before the 10-12 hr stabilizing period I will leave it alone. The reason being that it was off only 1 hour and its temp is already stable - the decreased current is the expected response to the increase in temp.

Will begin next set on original observation schedule. Set #11 Power failure again while

Rodger & I were attempting to wire up and start the generator in the event of a power failure. Too late Nator! System down again.

Generator now running, system powered up waiting to relock spring and laser. Generator will require refueling every 2 to 3 hours. Current PLAN is to operate until we have 18 sets (19)

as the data looks good - very good and the problems are many.

Generator Refueling schedule: 5:30P

8:00P

11 liter carosen / hr

10:30P

5 liter capacity

01:00A

03:30A

0600A

0830A

Next Page

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Nagrakot Observatory

March 31, 1991 Started Set # 13 @ ~ 1006Z on generator power. LASER is not appreciating this on/off activity relocking @ .220A and drifting down to .187A after 70 drops. This generator they've supplied us with is a piece of shit and almost dies every 10 minutes causing the chain to switch off and since the topaz's haven't recharged from this morning there not likely to hold the system for long. Also the generator is injecting a lot of noise into the data which in my opinion is very undesirable. The power has been off for over an hour now and if it doesn't come back on soon I'll be pissed. The First Blue Set run w/ the generator or had a considerably higher signal of 60 microgals but still a very reasonable value considering other sites we've visited during this trip. 1106Z Generator Died and UPS unable to maintain system. Everything shut off. Restarted it to recharge UPS but it died again. Thunderstorm has moved in and were calling it quits until the main power is returned. unplugged unit & ground to safeguard equipment. Setting up battery to run ion pump as were uncertain as to how long the UPS will run the pump.

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Napierkot Observatory contin
April 1, 1991 1400Z Began the 3rd restart.
Spent all day recharging batteries to keep
the Ion pump and GPS receivers up while
the power remained out. Power came up around
1800 local and I was able to begin obs.
Power capability at this site is at the very edge
of operational requirements. Example - Turn the
scope on to fix the fringe signal and the current
draw w/ EVERYTHING ON causes the UPS to trigger
as the power to the Elgar drops out. Fun and
games.

0100Z March April 2, 1991 Last 6 sets showed a
offset of ≈ 10 microgals from earlier sets which
following a 1 diameter adjustment of the Interferometer
vertically raised the red value by 7 microgals.
Difficulty in leveling the interferometer is I believe
related to the pinhole objective lens collimation
which was upset during cargo transport. Last
night's initial check revealed 2 reflected spots
and one stationary. I chose the brightest spot
to collimate but this morning that spot had
drifted. Sorry the data's not perfectly streamline
but I think that explains it.

0500Z Room temperature rising to high.
Inside 24°C outside 19°C - opened door.
0600Z Sat - Telephone Transmitter was left connected
by accident and seriously affected a number of Drops
on the 9th set (last 12 sets). Temperature is also very
high and the laser is at the edge of unlocking.
More activity than during the previous 20 sets is

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Organization NOAA/NOS/NGS

COMMENTS

Nagrakot Observatory Contin

Apparent on the SS trace marked by spikes.

only 3 more sets to go. Jack & I decided to shut system down w/ 21 sets under our belt.

The temperature became abnormally high and the laser was about to unlock.

System ✓ OK - still 1 extra reflected spot in the alcohol pool. Must check this with Glenn when we arrive in Hawaii and do a quick check of the collimation.

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ABSOLUTE GRAVITY STATION ORIENTATION DIAGRAM

STATION: NAGROKOT, NEPAL

DATE: MARCH 31, 1991

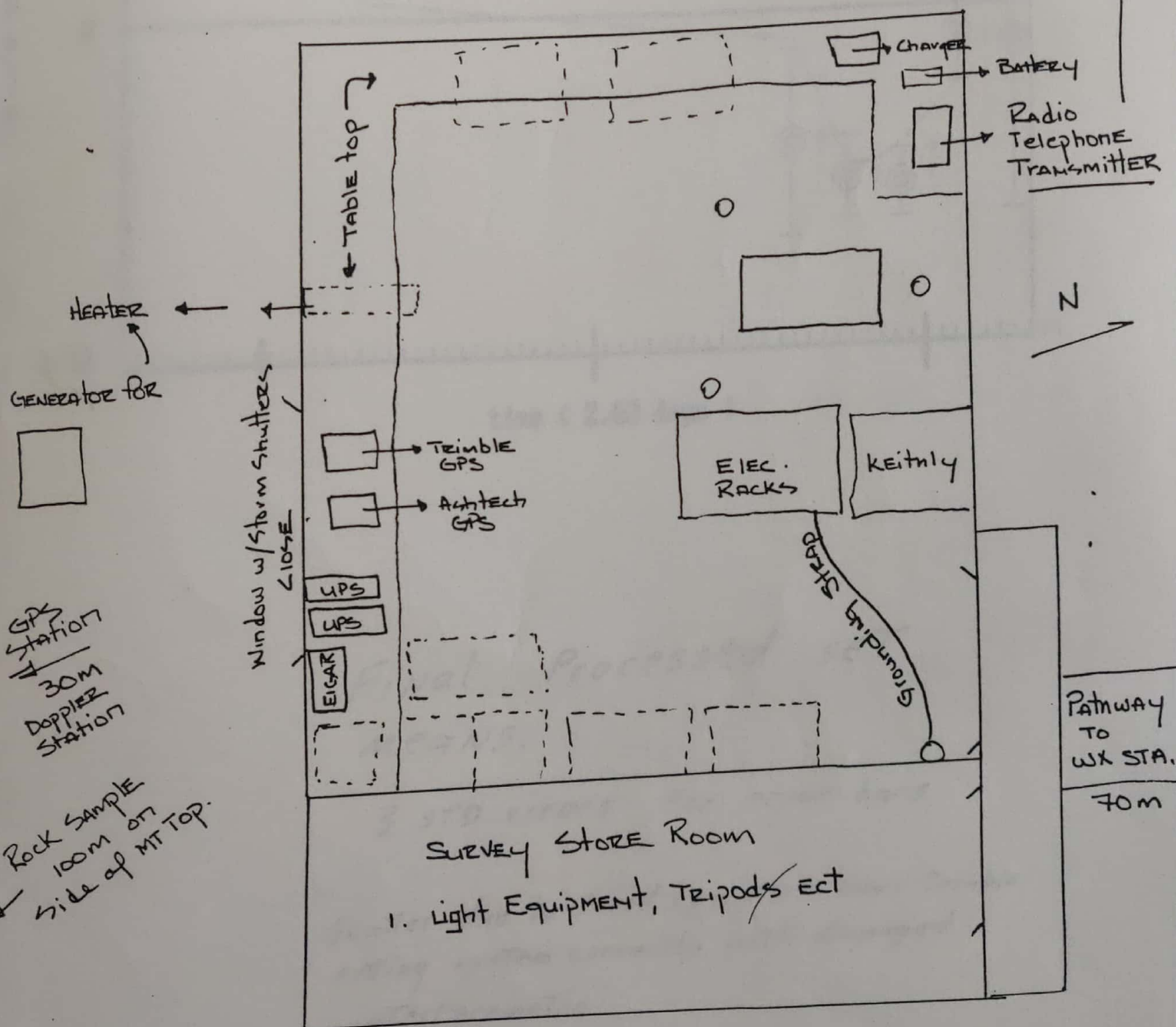
HMG SURVEY DEPT, NAGROKOT OBSERVATORY

Timing & Radio Room

Brick Building
12 inch concrete Pad
on Biotite Schist

TRIANGULATION
Station
Heli Pad, Gravity
Station

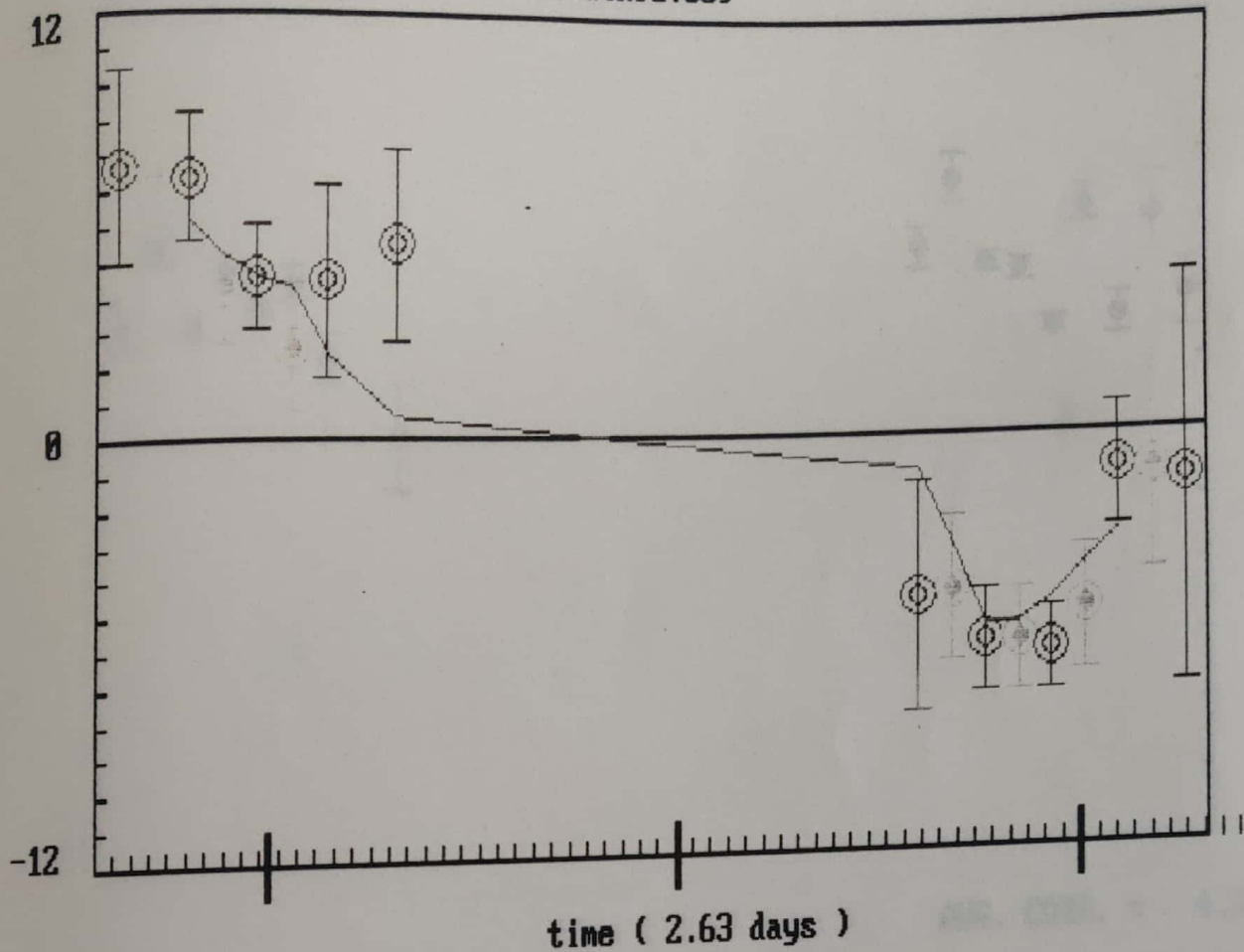
Disconnected During
OBSERVATIONS
by G-52



STATION CONTACT: MR. Buddhi N. Shrestha 977 1 411 897
DIRECTOR GENERAL HMG SURVEY DEPT

GRAVITY SET DISTRIBUTION

aakath91.089



Final Processed set
means.

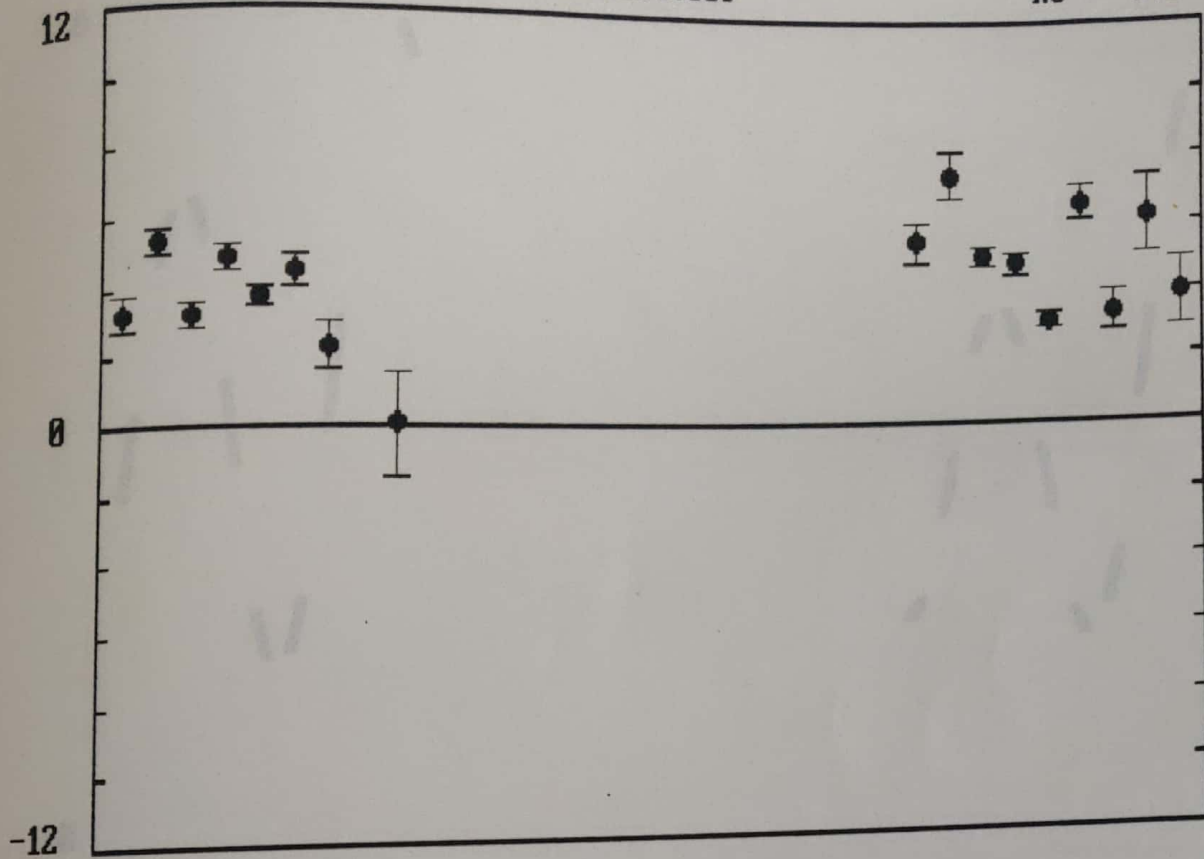
3 STD errors For error bars

Scatter due To Field operators have Trouble
setting system correctly with damaged
interFerometer.

BC = 5.7

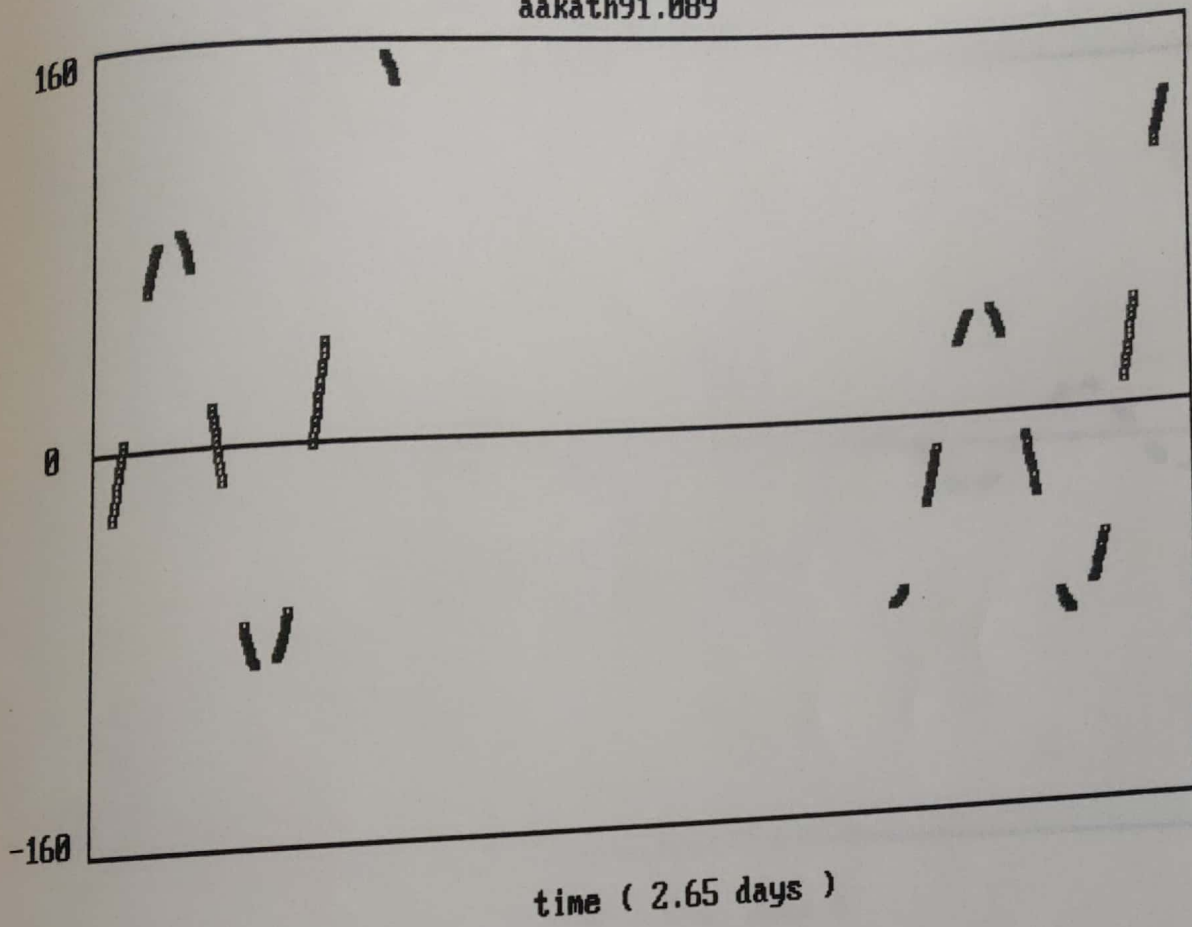
SET SYSTEM RESPONSE CORRECTION
aakath91.089

RC = 3.3

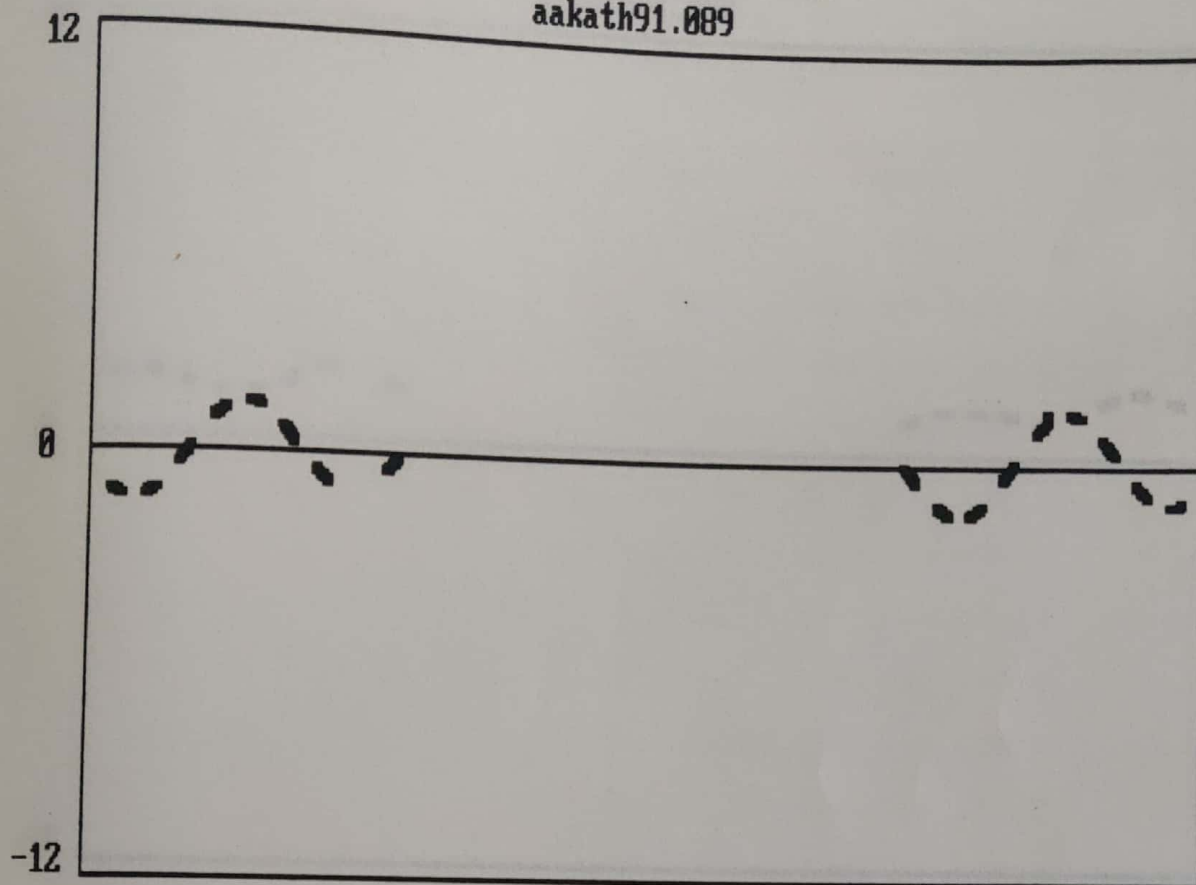


time (2.63 days) AUG. CORR. = 4.3

LUNAR-SOLAR TIDE CORRECTION
aakath91.089

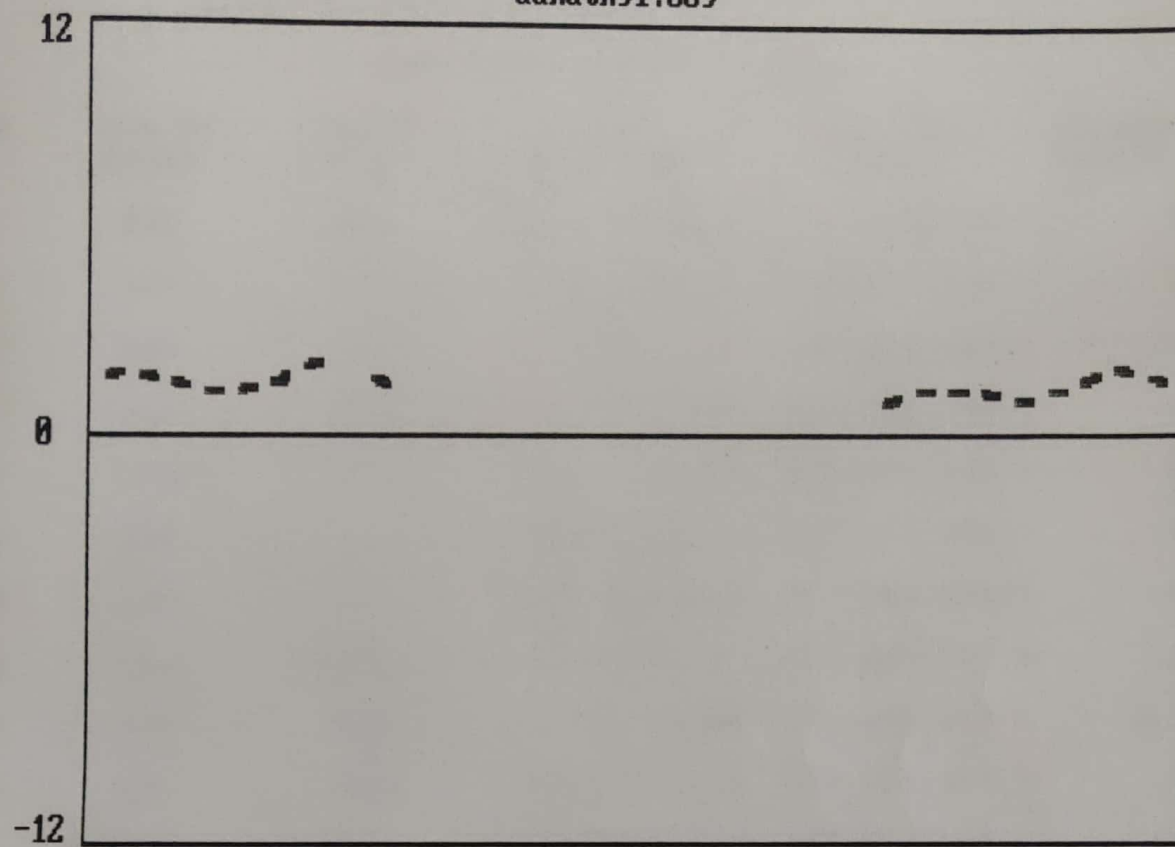


OCEAN LOADING CORRECTION
aakath91.089



time (2.65 days)

LOCAL ATMOSPHERIC MASS CORRECTION
aakath91.009



time (2.65 days)

NGS ABSOLUTE GRAVITY OBSERVATIONS From aakath91.089
 This drop set has been previously processed for:
 three sigma acceptance limit
 gravitational tide correction

DROP SET MEANS SUMMARY

drop set	num of drops	laser mode	mean date/time	mean grav (ugal)	sd mean (ugal)	sd obs (ugal)
# 1	250	RED	910330152056	978 494 428.8	.9	14.6
# 2	247	BLUE	910330172055	978 494 409.3	.5	8.6
# 3	249	RED	910330192056	978 494 427.7	.6	9.0
# 4	248	BLUE	910330212101	978 494 404.1	.5	8.1
# 5	249	RED	910330232055	978 494 423.6	.5	8.0
# 6	246	BLUE	910331012055	978 494 402.7	.6	9.7
# 7	248	RED	910331032056	978 494 424.7	.9	13.7
# 8	234	RED	910331072101	978 494 425.9	.9	13.6
# 9	234	RED	910401142106	978 494 416.2	1.1	16.1
# 10	237	BLUE	910401162101	978 494 397.7	.7	10.2
# 11	250	RED	910401182100	978 494 415.6	.5	8.2
# 12	248	BLUE	910401202055	978 494 395.1	.5	8.3
# 13	247	RED	910401222055	978 494 413.2	.4	6.8
# 14	236	BLUE	910402002100	978 494 394.3	.6	8.9
# 15	242	RED	910402022055	978 494 418.4	.6	9.7
# 16	246	BLUE	910402042110	978 494 399.9	1.0	16.2
# 17	215	RED	910402062136	978 494 419.8	2.0	28.6

10 dropsets weighted mean of red mode observations = 4 420.1 5.7
 7 dropsets weighted mean of blue mode observations = 4 400.7 5.3
 average of weighted red and blue means = 4 410.4 5.5

average standard deviation of observation = 11.7

NGS ABSOLUTE GRAVITY OBSERVATIONS From aakath91.089
 This drop set has been previously processed for:
 three sigma acceptance limit
 gravitational tide correction

DROP SET MEANS SUMMARY
 OFFSET CORRECTED

drop set	num of drops	laser mode	mean date/time	mean grav (ugal)	residual (ugal)
# 1	250	RED	910330152056	978 494 419.1	8.7
# 2	247	BLUE	910330172055	978 494 418.9	8.6
# 3	249	RED	910330192056	978 494 418.1	7.7
# 4	248	BLUE	910330212101	978 494 413.8	3.4
# 5	249	RED	910330232055	978 494 413.9	3.5
# 6	246	BLUE	910331012055	978 494 412.4	2.0
# 7	248	RED	910331032056	978 494 415.1	4.7
# 8	234	RED	910331072101	978 494 416.3	5.9
# 9	234	RED	910401142106	978 494 406.5	-3.9
# 10	237	BLUE	910401162101	978 494 407.4	-3.0
# 11	250	RED	910401182100	978 494 405.9	-4.5
# 12	248	BLUE	910401202055	978 494 404.8	-5.6
# 13	247	RED	910401222055	978 494 403.5	-6.9
# 14	236	BLUE	910402002100	978 494 404.0	-6.4
# 15	242	RED	910402022055	978 494 408.7	-1.7
# 16	246	BLUE	910402042110	978 494 409.6	-.8
# 17	215	RED	910402062136	978 494 410.1	-.3

average of weighted red and blue means = 4 410.4 s.d. mean = 5.5

average standard deviation of observation = 11.7

NGS ABSOLUTE GRAVITY OBSERVATIONS From aakath91.089
 This drop set has been previously processed for:
 gravitational tide correction

DROP SET MEANS SUMMARY

drop set	num of drops	laser mode	mean date/time	mean grav (ugal)	sd mean (ugal)	sd obs (ugal)
# 1	250	RED	910330152056	978 494 428.8	.9	14.6
# 2	250	BLUE	910330172055	978 494 409.6	.6	9.1
# 3	250	RED	910330192056	978 494 427.6	.6	9.2
# 4	250	BLUE	910330212101	978 494 404.3	.5	8.4
# 5	250	RED	910330232055	978 494 423.5	.5	8.2
# 6	250	BLUE	910331012055	978 494 402.8	.7	10.6
# 7	250	RED	910331032056	978 494 424.8	.9	14.6
# 8	250	RED	910331072101	978 494 425.2	2.0	32.4
# 9	250	RED	910401142106	978 494 413.0	2.4	38.0
# 10	250	BLUE	910401162101	978 494 397.8	2.5	39.4
# 11	250	RED	910401182100	978 494 415.6	.5	8.2
# 12	250	BLUE	910401202055	978 494 395.4	.5	8.7
# 13	250	RED	910401222055	978 494 412.9	.5	7.3
# 14	250	BLUE	910402002100	978 494 394.1	.8	12.5
# 15	250	RED	910402022055	978 494 417.9	.8	13.0
# 16	250	BLUE	910402042110	978 494 401.0	1.1	18.0
# 17	250	RED	910402062136	978 494 420.8	5.6	89.2

10 dropsets weighted mean of red mode observations = 4 419.9 6.0
 7 dropsets weighted mean of blue mode observations = 4 401.8 5.5
 average of weighted red and blue means = 4 410.9 5.8

average standard deviation of observation = 20.1

NGS ABSOLUTE GRAVITY OBSERVATIONS From aakath91.089
 This drop set has been previously processed for:
 gravitational tide correction

DROP SET MEANS SUMMARY
 OFFSET CORRECTED

drop set	num of drops	laser mode	mean date/time	mean grav (ugal)	residual (ugal)
# 1	250	RED	910330152056	978 494 419.7	8.8
# 2	250	BLUE	910330172055	978 494 418.7	7.8
# 3	250	RED	910330192056	978 494 418.6	7.7
# 4	250	BLUE	910330212101	978 494 413.4	2.6
# 5	250	RED	910330232055	978 494 414.4	3.6
# 6	250	BLUE	910331012055	978 494 411.8	1.0
# 7	250	RED	910331032056	978 494 415.7	4.9
# 8	250	RED	910331072101	978 494 416.1	5.2
# 9	250	RED	910401142106	978 494 403.9	-6.9
# 10	250	BLUE	910401162101	978 494 406.9	-4.0
# 11	250	RED	910401182100	978 494 406.5	-4.3
# 12	250	BLUE	910401202055	978 494 404.5	-6.4
# 13	250	RED	910401222055	978 494 403.8	-7.0
# 14	250	BLUE	910402002100	978 494 403.2	-7.6
# 15	250	RED	910402022055	978 494 408.8	-2.0
# 16	250	BLUE	910402042110	978 494 410.1	-.8
# 17	250	RED	910402062136	978 494 411.8	.9

average of weighted red and blue means = 4 410.9 s.d. mean = 5.8

average standard deviation of observation = 20.1

NGS ABSOLUTE GRAVITY OBSERVATIONS From aakath91.089
 This drop set has been previously processed for:
 three sigma acceptance limit
 gravitational tide correction
 local atmospheric pressure correction

DROP SET MEANS SUMMARY

drop set	num of drops	laser mode	mean date/time	mean grav (ugal)	sd mean (ugal)	sd obs (ugal)
# 1	250	RED	910330152056	978 494 430.6	.9	14.6
# 2	247	BLUE	910330172055	978 494 411.0	.5	8.6
# 3	249	RED	910330192056	978 494 429.3	.6	9.0
# 4	248	BLUE	910330212101	978 494 405.4	.5	8.1
# 5	249	RED	910330232055	978 494 425.0	.5	8.0
# 6	246	BLUE	910331012055	978 494 404.4	.6	9.7
# 7	248	RED	910331032056	978 494 426.8	.9	13.7
# 8	234	RED	910331072101	978 494 427.6	.9	13.6
# 9	234	RED	910401142106	978 494 417.2	1.1	16.1
# 10	237	BLUE	910401162101	978 494 399.0	.7	10.2
# 11	250	RED	910401182100	978 494 416.9	.5	8.2
# 12	248	BLUE	910401202055	978 494 396.3	.5	8.3
# 13	247	RED	910401222055	978 494 414.2	.4	6.8
# 14	236	BLUE	910402002100	978 494 395.6	.6	8.9
# 15	242	RED	910402022055	978 494 420.1	.6	9.7
# 16	246	BLUE	910402042110	978 494 401.8	1.0	16.2
# 17	215	RED	910402062136	978 494 421.4	2.0	28.6

10 dropsets weighted mean of red mode observations = 4 421.5 5.9
 7 dropsets weighted mean of blue mode observations = 4 402.1 5.5
 average of weighted red and blue means = 4 411.8 5.7

average standard deviation of observation = 11.7

NGS ABSOLUTE GRAVITY OBSERVATIONS From aakath91.089
 This drop set has been previously processed for:
 three sigma acceptance limit
 gravitational tide correction
 local atmospheric pressure correction

DROP SET MEANS SUMMARY
 OFFSET CORRECTED

drop set	num of drops	laser mode	mean date/time	mean grav (ugal)	residual (ugal)
# 1	250	RED	910330152056	978 494 420.9	9.1
# 2	247	BLUE	910330172055	978 494 420.7	8.9
# 3	249	RED	910330192056	978 494 419.6	7.8
# 4	248	BLUE	910330212101	978 494 415.1	3.3
# 5	249	RED	910330232055	978 494 415.3	3.5
# 6	246	BLUE	910331012055	978 494 414.0	2.2
# 7	248	RED	910331032056	978 494 417.2	5.4
# 8	234	RED	910331072101	978 494 417.9	6.1
# 9	234	RED	910401142106	978 494 407.5	-4.3
# 10	237	BLUE	910401162101	978 494 408.6	-3.2
# 11	250	RED	910401182100	978 494 407.2	-4.6
# 12	248	BLUE	910401202055	978 494 406.0	-5.8
# 13	247	RED	910401222055	978 494 404.5	-7.3
# 14	236	BLUE	910402002100	978 494 405.3	-6.6
# 15	242	RED	910402022055	978 494 410.4	-1.4
# 16	246	BLUE	910402042110	978 494 411.5	-.3
# 17	215	RED	910402062136	978 494 411.8	.0

average of weighted red and blue means = 4 411.8 s.d. mean = 5.7
 average standard deviation of observation = 11.7

NGS ABSOLUTE GRAVITY OBSERVATIONS From aakath91.089
 This drop set has been previously processed for:
 three sigma acceptance limit
 gravitational tide correction
 local atmospheric pressure correction
 ocean loading correction

DROP SET MEANS SUMMARY

drop set	num of drops	laser mode	mean date/time	mean grav (ugal)	sd mean (ugal)	sd obs (ugal)
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# 2	247	BLUE	910330172055	978 494 409.8	.5	8.6
# 3	249	RED	910330192056	978 494 429.1	.6	9.0
# 4	248	BLUE	910330212101	978 494 406.5	.5	8.1
# 5	249	RED	910330232055	978 494 426.3	.5	8.0
# 6	246	BLUE	910331012055	978 494 404.9	.6	9.7
# 7	248	RED	910331032056	978 494 426.2	.9	13.7
# 8	234	RED	910331072101	978 494 427.2	.9	13.6
# 9	234	RED	910401142106	978 494 417.0	1.1	16.1
# 10	237	BLUE	910401162101	978 494 397.7	.7	10.2
# 11	250	RED	910401182100	978 494 415.7	.5	8.2
# 12	248	BLUE	910401202055	978 494 396.2	.5	8.3
# 13	247	RED	910401222055	978 494 415.4	.4	6.8
# 14	236	BLUE	910402002100	978 494 397.1	.6	8.9
# 15	242	RED	910402022055	978 494 420.7	.6	9.7
# 16	246	BLUE	910402042110	978 494 401.2	1.0	16.2
# 17	215	RED	910402062136	978 494 420.3	2.0	28.6

10 dropsets weighted mean of red mode observations = 4 421.7 5.7
 7 dropsets weighted mean of blue mode observations = 4 402.3 5.3
 average of weighted red and blue means = 4 412.0 5.5

average standard deviation of observation = 11.7

NGS ABSOLUTE GRAVITY OBSERVATIONS From aakath91.089
 This drop set has been previously processed for:
 three sigma acceptance limit
 gravitational tide correction
 local atmospheric pressure correction
 ocean loading correction

DROP SET MEANS SUMMARY
 OFFSET CORRECTED

drop set	num of drops	laser mode	mean date/time	mean grav (ugal)	residual (ugal)
# 1	250	RED	910330152056	978 494 419.6	7.7
# 2	247	BLUE	910330172055	978 494 419.5	7.6
# 3	249	RED	910330192056	978 494 419.4	7.4
# 4	248	BLUE	910330212101	978 494 416.2	4.3
# 5	249	RED	910330232055	978 494 416.6	4.6
# 6	246	BLUE	910331012055	978 494 414.6	2.6
# 7	248	RED	910331032056	978 494 416.5	4.5
# 8	234	RED	910331072101	978 494 417.5	5.5
# 9	234	RED	910401142106	978 494 407.2	-4.7
# 10	237	BLUE	910401162101	978 494 407.4	-4.5
# 11	250	RED	910401182100	978 494 405.9	-6.0
# 12	248	BLUE	910401202055	978 494 406.0	-6.0
# 13	247	RED	910401222055	978 494 405.7	-6.3
# 14	236	BLUE	910402002100	978 494 406.8	-5.1
# 15	242	RED	910402022055	978 494 411.0	-1.0
# 16	246	BLUE	910402042110	978 494 410.9	-1.1
# 17	215	RED	910402062136	978 494 410.6	-1.4

average of weighted red and blue means = 4 412.0 s.d. mean = 5.5
 average standard deviation of observation = 11.7

ABSOLUTE GRAVITY: A RECONNAISSANCE TOOL FOR STUDYING VERTICAL CRUSTAL MOTIONS

T. M. Niebauer, J. K. Hoskins,¹ and J. E. Faller

Joint Institute for Laboratory Astrophysics, National Bureau of Standards
and University of Colorado, Boulder

Abstract. A major effort is under way to develop highly portable absolute gravimeters having an ultimate accuracy of 3–5 μGal , an accuracy which translates into a height sensitivity of several centimeters. We are just finishing the construction of six such units. Measurements at the Joint Institute for Laboratory Astrophysics with one of these new instruments agree well with the earlier measurements made in 1981 and 1982 with a previous generation instrument. Recent measurements at the International Bureau of Weights and Measures in Sevres, France, as a part of an international intercomparison of absolute gravimeters, also show good agreement with the other instruments.

Measurement of the absolute value of the free-fall acceleration "g" has long been a matter of scientific interest. Present-day methods of measuring the absolute value of g employ ballistic systems involving either direct free-fall or symmetrical rise-and-fall methods. The earliest such measurements employed the direct free-fall method and geometrical optics to determine the position of the dropped object as a function of time. More recently, laser interferometry has been used almost exclusively.

A major effort to develop a new generation of high-precision absolute gravimeters is in the final stages at the Joint Institute for Laboratory Astrophysics (JILA) located at the University of Colorado in Boulder, Colorado. These gravimeters interferometrically measure the position of a free-falling object as a function of time and thereby permit the determination of the free-fall acceleration. This paper will discuss the use of absolute gravity for the study of vertical motions, the status of the JILA absolute gravity instruments, and the advantages and near-term prospects of using them for this purpose.

Traditionally, vertical height information has been derived mainly from leveling data. However, even using automated leveling systems, the cost per kilometer is high, from \$350/km to rerun an existing line to between \$500 and \$600/km to run a new line (G. J. Mitchell, private communication, 1986). A number of extraterrestrial techniques and systems also exist for measuring vertical movements of the earth's surface such as laser satellite ranging, very long baseline interferometry, and using ground receivers together

with the NAVSTAR global positioning system satellites. These methods are now capable of achieving the interesting accuracies of between 1 and 3 cm and are therefore likely to play an increasingly important role in determining vertical motions. Their costs are still high; but these costs, particularly those associated with the global positioning satellite system approach, should soon be lowered.

Gravity measurements, both relative and absolute, given sufficient measurement precision, provide a comparatively inexpensive way to look for vertical crustal movements. A 1-cm vertical crustal motion would result in a gravity change of approximately 3 μGal were no change in the local mass distribution to occur. The actual change in gravity observed in connection with a 1-cm vertical displacement will generally be 2–3 μGal but can be outside this range for some crustal movement mechanisms [Jachens, 1978a,b]. To differentiate, however, between subsurface density changes and vertical height changes, one must use one of the geometrical geodetic systems. Gravity does, however, provide an excellent and low cost reconnaissance tool with which to gather large amounts of preliminary data which then, for those areas in which gravity changes are occurring, can be checked and interpreted in combination with the other (geometrical) vertical data. If vertical motions are subsequently confirmed by other means, the observed gravity changes can help to determine the mechanism responsible for the motions.

In using gravity measurements as a reconnaissance tool to look for vertical movements, absolute gravity measurements have a number of advantages over relative gravity measurements: the most important of these being that absolute gravity is a "point technique." A single measurement produces a gravity value, in some sense a measure of the distance from the center of the earth, which depends only on the basic standards of length and time. Relative gravity measurements (as well as conventional leveling techniques) must necessarily be tied to a (presumed) stable external reference point which complicates the measurement process and inevitably raises questions about the stability of that reference point over the appropriate time frame. In a relative gravimeter (see, for example Clark [1984]), the spring, whose length is essentially the measured parameter, displays secular creep as well as episodic changes in its length. Vibrations encountered while transporting these devices and stresses due to clamping only serve to exacerbate these problems. In addition, nonlinearities in the adjusting screw and its associated lever reduction mechanism have to be carefully calibrated if their effects are to be removed. In practice, the measurement precision depends on the par-

¹Now at Northrup Corporation, Hawthorne, California.

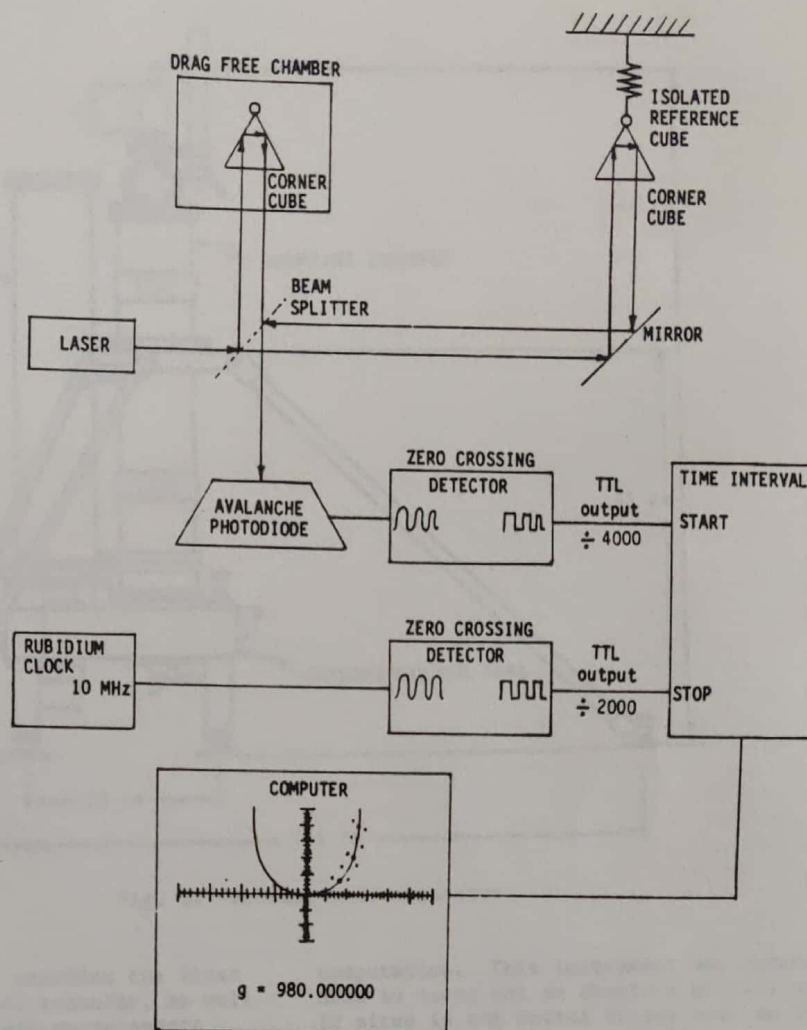


Fig. 1. Block diagram of free-fall method.

ticular instrument used, the station-to-station distance, and also on the gravity difference. Without special precautions, relative gravimeters typically reach precisions of between ± 30 and ± 100 μGal for a single measurement of a given difference in gravity. Extreme care is required to reduce this error to the ± 5 to ± 10 μGal range [Torge, 1985].

By contrast, the accuracy of absolute free-fall instruments depends mainly on the reproducibility of the basic standards of length and time, and a stabilized laser provides the length standard and an atomic (rubidium) clock provides the time standard. The absolute wavelength of the laser and the frequency of the atomic clock can easily be measured directly in the laboratory. The drifts in these "standards" are low enough so that they can be used for months with negligible error contributions at the parts in 10^9 level of accuracy. Further, these "standards" are less subject to the ordinary vibration in transit, environmental temperature, etc., problems which have proven difficult with traditional relative gravimeters at the microGal level of sensitivity.

Modern-day absolute gravity instruments have been developed and improved over the past 30 years through the utilization of available

technology. In practice, they all measure the position of a freely falling mass as a function of time (with exquisite sensitivity) and from that motion determine the value of g (Figure 1). Two types of free-fall instruments have been developed: the first utilizes simple free fall, and the second uses an up-and-down trajectory [Faller and Sakuma, 1986]. In each case, g is determined by fitting a quadratic expression to the measured trajectory. In practice, a Michelson-type laser interferometer is used to sense the position of the falling object during its fall. The dropped object contains a cube corner (a special type of optical mirror that reflects the laser directly back, independent of the cube's exact orientation). The occurrence times of the zero crossing of the fringes then provide the necessary information with which to calculate g .

The first laser interferometric g measurements were made in 1962 by J. E. Faller using an early commercially available He-Ne laser in what had been designed as a white-light-fringe g apparatus. The first portable laser interferometer absolute gravimeter was developed by J. A. Hammond and J. E. Faller at JILA and Wesleyan University with support from the Air Force Geophysics Laboratory (AFGL). With this apparatus, which had an accuracy of 50 μGal , data were taken at eight

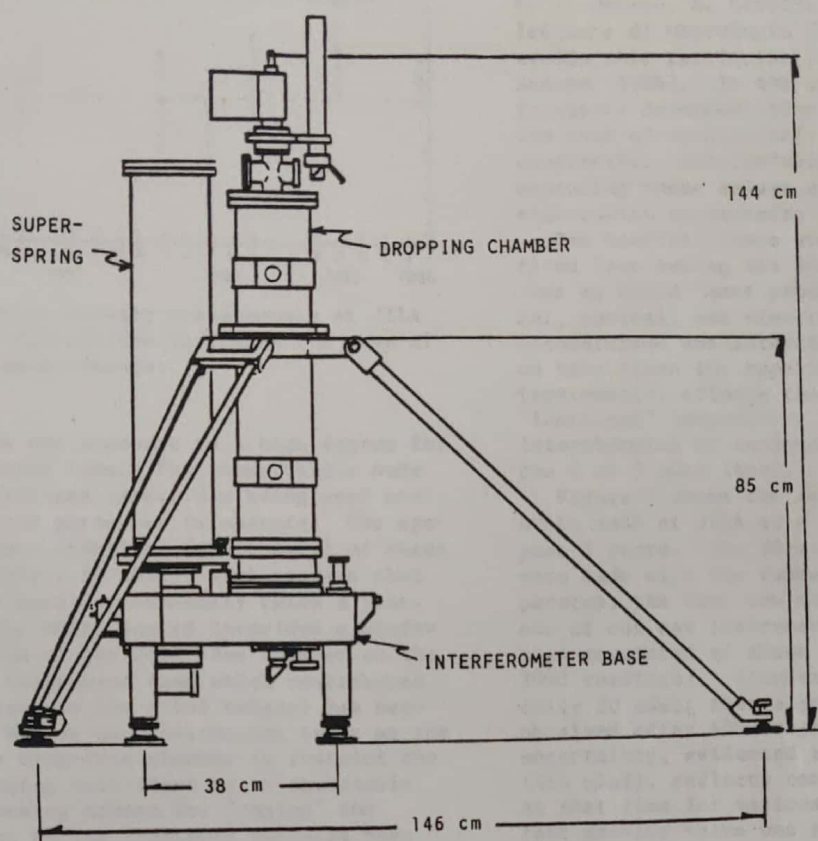


Fig. 2. Schematic of apparatus.

different sites. These data provided the first transatlantic absolute gravity transfer, as well as the first series of absolute measurements along the American calibration line [Hammond and Faller, 1971]. Although successfully fulfilling its intended goals, the apparatus was quite heavy (880 kg), and the better part of a week was required to set it up and complete a measurement.

A modification of the Hammond-Faller gravity apparatus was later made by J. A. Hammond, R. Sands, and R. Iliff at AFGL. The result was a somewhat more compact instrument which also incorporated other changes that made it easier to set up. This modified apparatus required 1 or 2 days at each site and its accuracy was 10–15 μGal [Iliff and Sands, 1983].

A third generation laser interferometer apparatus was built by M. A. Zumberge and J. E. Faller at JILA. This instrument included several fundamental changes, the use of a "drag-free" chamber and a "super spring," which brought its accuracy into the 6–10 μGal range. Particular emphasis was placed on decreasing its weight and the time required to set it up. Its free-fall inside a "drag-free" chamber approach provided both a less stringent vacuum requirement and a faster duty cycle. The use of a long-period "super spring" isolation device [Rinker and Faller, 1984] served to reduce the effect of microseisms and man-made vibrations as well as to isolate the apparatus from a possible systematic recoil associated with the release of the dropped object. Finally, the digital design permitted real-time computer readings of both fringe and time information, thus resulting in a faster g

computation. This instrument was successfully used to carry out an absolute gravity survey at 12 sites in the United States over an 8-week time period [Zumberge et al., 1983].

Motivated by the potential uses, we have almost completed construction of six absolute gravity meters at JILA. These instruments employ the method of free-fall, as did their predecessors, and consist of four basic parts: a "drag-free" dropping chamber, a long-period isolation device, a stabilized laser, and the necessary timing electronics. They have been designed to be easily portable and usable in a variety of field situations, and their ultimate accuracy is expected to be of the order of 3–5 μGal .

In designing the latest group of JILA instruments (Figure 2), special emphasis has been placed on making them more durable and easy to use in the field. The mechanical integrity and stability have been improved; for example, collet-type drive clamps are used to eliminate slipping and backlash. Components most subject to wear, such as the vee grooves and balls that serve to kinematically locate the dropped object prior to its release, are now easily replaced. The new interferometer base is much easier to use (the optical components are on the sides rather than on the bottom) and takes substantially less time (minutes as opposed to the order of a half-hour or more) to set up than did the earlier ones. A new optical system for setting up and aligning the interferometer and dropping chamber has been built into the system. A transmission (rather than reflection) method of autocollimation permits the use of an oil rather than a Hg vertical

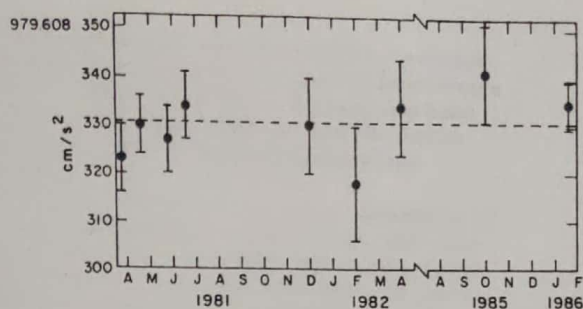


Fig. 3. Absolute gravity measurements at JILA site AA. The dotted line indicates the mean of all of these measurements.

reference pool and accounts to a high degree for the reduced setup time. The commercially made frequency stabilized lasers now being used are simple for field personnel to operate. The specified frequency stability (drift rate) of these lasers (± 2 MHz/yr, ± 4 μ Gal), will require that they be calibrated approximately twice a year.

The pellicle which sealed (provided a window for) the bottom of the drag-free chamber on the earlier JILA instrument (and which contributed the largest term to the error budget) has been eliminated. We now use collimating tubes on the bottom of the drag-free chamber to restrict the momentum-changing collisions to an acceptable number. A locking scheme for "caging" the dropped object during transport has also been devised and built into the new instruments. The redesigned tripod for the dropping chamber, while mechanically stiff, has the ability to accommodate itself to many different pier shapes and sizes.

All of the electronics circuits (dropping chamber servo, super spring servo, fringe detecting, and scaling) have been placed on printed circuit boards, thus facilitating the process of finding and replacing faulty circuits. The timing electronics has been reduced from three rack-mounted boards to only one. A 50-MHz bandwidth avalanche photodiode has replaced the (fragile) photomultiplier which was used in earlier instruments.

The quality of the zero-crossing detection electronics has been recognized as absolutely crucial to the success of the straight free-fall type of measurement. A systematic error may be introduced if a frequency dependent time delay exists, since this alters the observed transit time of each fringe, resulting in an apparent change in g . For example, consider a time delay of the form $D(f) = A + B/f + C*f$, where D is the time delay and f is the frequency. On substituting this into the equation $X = 0.5*g*(T+D)^2$, one finds that the constant term "changes" the initial velocity (no problem) and that the term inversely proportional to frequency "shifts" the initial position (again no problem) but that the linear term results in an erroneous acceleration and therefore a wrong g value. Although relatively large time delays of the first two types will not change the measured value for g , to keep from affecting g at the microGal level, one must strenuously avoid delays of the last type. The up-and-down approach used by A. Sakuma at the Bureau International des Poids et Mesures, and

by I. Marson, G. Cerutti, and L. Cannizzo at the Istituto di Metrologia G. Colonnetti, Torino, avoids this particular problem [Faller and Sakuma, 1986]. In the up-and-down method, some frequency dependent time delays cancel but at the cost of considerably greater mechanical complexity. Considerable time has been spent measuring these delays and designing fast electronics to minimize this problem.

One benefit (there are some nonbenefits) derived from making six identical instruments is that we could "mass produce" parts. The mechanical, optical, and electronic components are standardized and interchangeable. Accordingly, we have taken the opportunity to look for any (systematic) effects that vary when changing "identical" components. To date, all of these interchanging of components effects have been at the 2 or 3 μ Gal level.

Figure 3 shows the absolute gravity measurements made at JILA as a function of time over the past 5 years. The first seven measurements shown were made with the Zumbege-Faller prototype apparatus; the last two measurements were made with one of our new instruments. The first determination consisted of about 2 days of data (around 3000 readings). Standard deviations are typically 20 μ Gal; thus a precision of 1 μ Gal is obtained after 400 drops (15 min). The large uncertainty, evidenced by the large error bar (± 10 μ Gal), reflects conservative allowances made at that time for various systematic errors. The last gravity value was obtained from a one month data set consisting of about 100,000 drops. A major contributor to the uncertainty (± 5 μ Gal) associated with this determination was the variations in air pressure and thus changes in the overhead (attractive) mass during the measurement process. We are now in the process of making air pressure corrections on this data set. Although this study is not yet concluded, correcting for the local air pressure yields an rms amplitude for the gravity residuals over this month of data of about 2 μ Gal. As is evident from the decrease in uncertainty between the two measurements with our new instrument, our study of possible systematic error sources is proceeding satisfactorily. The values given are referenced to 1-m height above the floor.

In June-July 1985 we took one of our new instruments to the International Bureau of Weights and Measures (BIPM) in Sevres, France, to participate in the Second International Comparison of Absolute Gravimeters. Nearly simultaneous measurements were made at six different sites with instruments from five different countries (United States, China, France, Italy, and Russia). The final and definitive result of this intercomparison awaits the least squares reduction of all of the relative ties which were made with 14 LaCoste Romberg relative gravimeters by the participants. Our result at a height of one meter for BIPM point A-5 is $980,926,325 \pm 9$ μ Gal. This value is the result of 4000 drops and uses a gravity gradient of 252 μ Gal/m to obtain the value at 1-m height from our instrument's 0.83-m effective measurement height. A preliminary intercomparison result (transferring all measurements to the same site) gives a value for the average of all measurements which differs from our transferred value by 0.7 μ Gal. Our value differs

from the average which excludes three (perhaps statistically meaningful) outlying measurements by 1.6 μGal . Details of this intercomparison will ultimately be published in the Bulletin d'Information of the Bureau Gravimetric International.

We are now at the point of final assembly of the last two instruments. The design and construction of the six new instruments has taken far longer than we first anticipated, in part because we were unwilling to compromise on field usability, operating convenience, or accuracy. In ensuing field applications (involving geodetic, geophysical, geological, tidal, and tectonic studies) we believe these absolute meters, which can easily be transported in a small van, will, among other things, provide a useful and reliable reconnaissance and measuring tool for studying vertical crustal motions.

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